

Learning points (top or bottom box)

- PML can present in patients without obvious immunosuppression and should be considered in progressive focal deficits with compatible MRI.
- Pembrolizumab may stabilise biopsy-proven PML after an initial phase of clinical and radiological worsening.
- Early diagnosis and carefully selected use of PD-1 blockade can help prevent further deterioration, though reversal of established damage is uncommon.

Background (left column)

- PML is a JC virus-mediated demyelinating disease of the CNS, classically seen in markedly immunocompromised patients.
- Management typically focuses on restoring immune function (e.g., ART in HIV, stopping immunosuppressive agents), but options are limited when this is not feasible and prognosis is poor.
- PD-1 inhibitors such as pembrolizumab can enhance JC virus-specific T-cell responses; small series show mixed but promising outcomes.
- We describe biopsy-proven PML in an immunocompetent host managed with pembrolizumab.

Case presentation (left column)

- A 53-year-old man with no prior comorbidities developed 15 days of progressive left lower-limb weakness.
- Symptoms evolved from a subtle limp to difficulty rising from a chair and inability to keep slippers on the left foot.
- There was no history of HIV, malignancy, or exposure to immunosuppressive or immunomodulatory drugs.
- Baseline MRI brain showed focal T2/FLAIR hyperintensity in the right para-central lobule white matter without contrast enhancement or perfusion abnormality.

Investigations (middle column)

- Autoimmune and inflammatory work-up including ANA/ENA, NMO/MOG antibodies, paired serum/CSF oligoclonal bands, and serum ACE levels was unremarkable.
- Viral screening for common neurotropic and systemic infections was negative.

- CSF analysis showed no leukocytes with normal protein and glucose; JCV PCR was negative despite later histological confirmation.
- HLA-B52 was positive, but there were no clinical or laboratory features of systemic vasculitis or connective-tissue disease.
- Immune profiling for primary immunodeficiency (beta-2 glycoprotein panel, CD4/CD8 counts, immunoglobulin levels) was within normal range.
- Stereotactic cortical biopsy revealed demyelination with SV40-positive inclusions in oligodendrocytes, establishing JC virus–associated PML

Treatment and clinical/radiological course (middle column)

- The patient initially received IV methylprednisolone for suspected demyelination but experienced continued progression of weakness.
- Given biopsy-proven PML with no reversible immunosuppression, pembrolizumab 2 mg/kg IV (150 mg) was started in September 2023.
- The planned second dose at 3 weeks was declined; by 3 weeks there was clinical deterioration and MRI showed enlargement of the right para-central lesion with a new non-enhancing right cerebellar lesion.
- Follow-up MRIs at 3 and 6 months demonstrated stabilisation of both lesions, and neurological deficits plateaued without further significant decline.
- A second pembrolizumab infusion was administered in May 2024; MRI 2 months later showed a mild reduction in the cerebellar lesion size, with the para-central lesion unchanged but non-enhancing.
- At 1-year follow-up, MRI appearances remained stable and no immune checkpoint inhibitor–related adverse events were observed.

Imaging figure box (right column)

Title above image

- Figure 1. Serial MRI over 1 year

Legend below composite

- To: Axial FLAIR MRI shows a hyperintense lesion in the right para-central lobule white matter without contrast enhancement.
- 3 weeks after the first pembrolizumab infusion: the para-central lesion has increased in size and a new non-enhancing lesion appears in the right cerebellum.
- 3- and 6-month follow-up scans: no further growth; both lesions remain non-enhancing and radiologically stable.

- 1-year follow-up: slight reduction of the right cerebellar lesion, with the larger para-central lesion unchanged in size and signal.

Discussion (right column, under MRI)

- Immune checkpoint inhibitors such as pembrolizumab and nivolumab target the PD-1 pathway, augmenting antiviral T-cell responses against JC virus.
- Case series report heterogeneous outcomes: some patients improve or stabilise with decreased CSF JCV load, whereas others show no benefit or succumb to progressive disease.
- Our patient, without evident immunosuppression, demonstrated early clinical and radiological worsening within 3 weeks of PD-1 blockade followed by long-term stabilisation, mirroring patterns of “pseudo-progression” described in prior cohorts.
- These findings support the concept that PD-1 inhibition may primarily halt further demyelination rather than reverse established white-matter damage, reinforcing the need for early recognition and intervention.
- Factors such as baseline disability, viral burden, T-cell exhaustion states, and concurrent immunosuppressive therapies likely influence response and should inform patient selection.
- Larger, standardised studies are unlikely due to PML’s rarity, but accumulating case data suggest pembrolizumab is a reasonable option when conventional immune reconstitution strategies are unavailable

Take-home messages (bottom box)

- Consider PML in patients with subacute focal neurological deficits and characteristic MRI changes even when standard immunosuppressive risk factors are absent.
- Pembrolizumab can stabilise biopsy-proven PML after an initial phase of clinical and radiological worsening, with durable disease control in selected patients.
- Early diagnosis, careful immune evaluation, and realistic counselling about expected stabilisation rather than reversal are essential when using PD-1 blockade in PML.